ECE343 Spring 2025

Objective 7.3: Ampere's Law



UNITED STATES AIR FORCE ACADEMY

Lesson	Topics	Reading	Notes
24	BVP (Capacitor, Coax), Electric Potential	5.16	LO 5.3
25	Electric Boundary Conditions	5.18-5.22	LO 5.4
26	BC's/Capacitance	5.23-5.24	
27	Electrostatic Lab	Lab packet	
Chapter 6: Steady Current & Conductivity			
28	Current density, Conductivity, Resistance	Ch. 6	
Chapter 7: Magnetostatics			
29	Magnetic Fields, Gauss' Law for Magnetism	7.1-7.3	LO 7.1-7.2 **GR 2**
30	Ampere's Law	7.4-7.5, 7.9	LO 7.3
31	Ampere's Law Applied	7.6-7.8	LO 7.3
32	Magnetic Boundary Conditions	7.10-7.14	LO 7.4
33	Magnetostatics Lab	Lab packet	
Chapter 8: Time-Varying Fields			
34	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.1
35	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.2-8.3
Chapter 9: Plane Waves in Lossless Media			
36	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
37	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
38	Uniform Plane Waves & Polarization	9.4-9.6	LO 9.2-9.3
39	Normal Incidence	Vol 2, 5.6-5.7	LO 9.4-9.5
40	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
41	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
42	Incidence Lab	TBD	LO 9.5

Ch 7 Learning Objectives

7.1 I can use the Biot-Savart law to simple magnetostatic systems to find the magnetic field at an observation point.

7.2 I can articulate the meaning of both the integral form and point form of Gauss' Law for magnetism.

7.3 I can use Ampere's Law to calculate the magnetic field at an observation point for simple magnetostatic systems.

7.4 I can apply the magnetic field boundary conditions to calculate magnetic fields on both sides of a material boundary.

Ampere's Law

Integral Form

$$\oint_C \vec{H} \cdot d\ell = I_{\text{encl}}$$

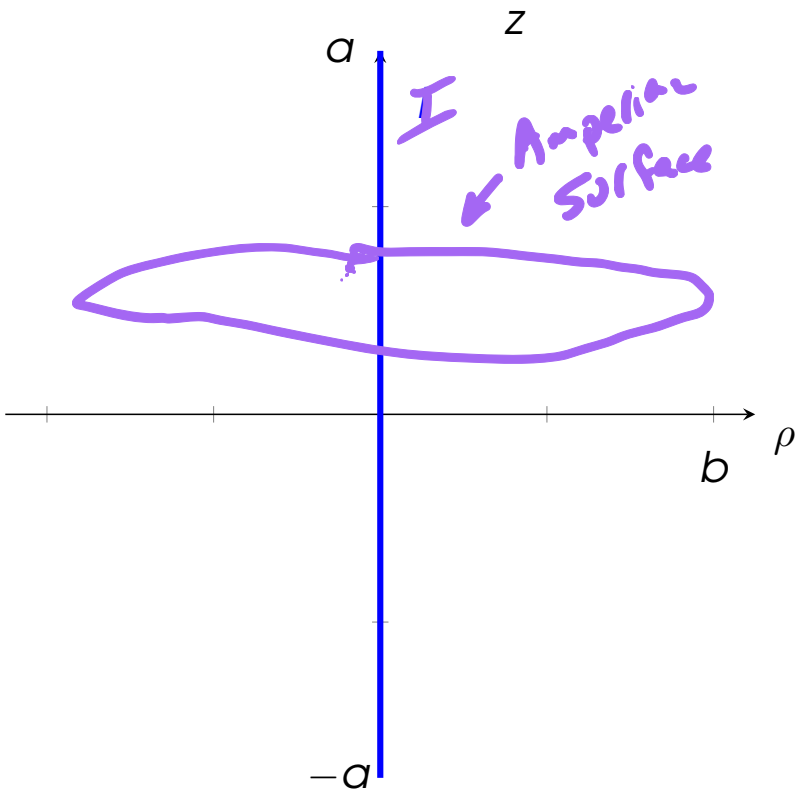
Point Form

$$\nabla \times \vec{H} = \vec{J}$$

$\vec{B}$ → Mag. flux density

AMPERE'S LAW: LINE CHARGE

Find $\vec{H}$ at a distance b from the LC.

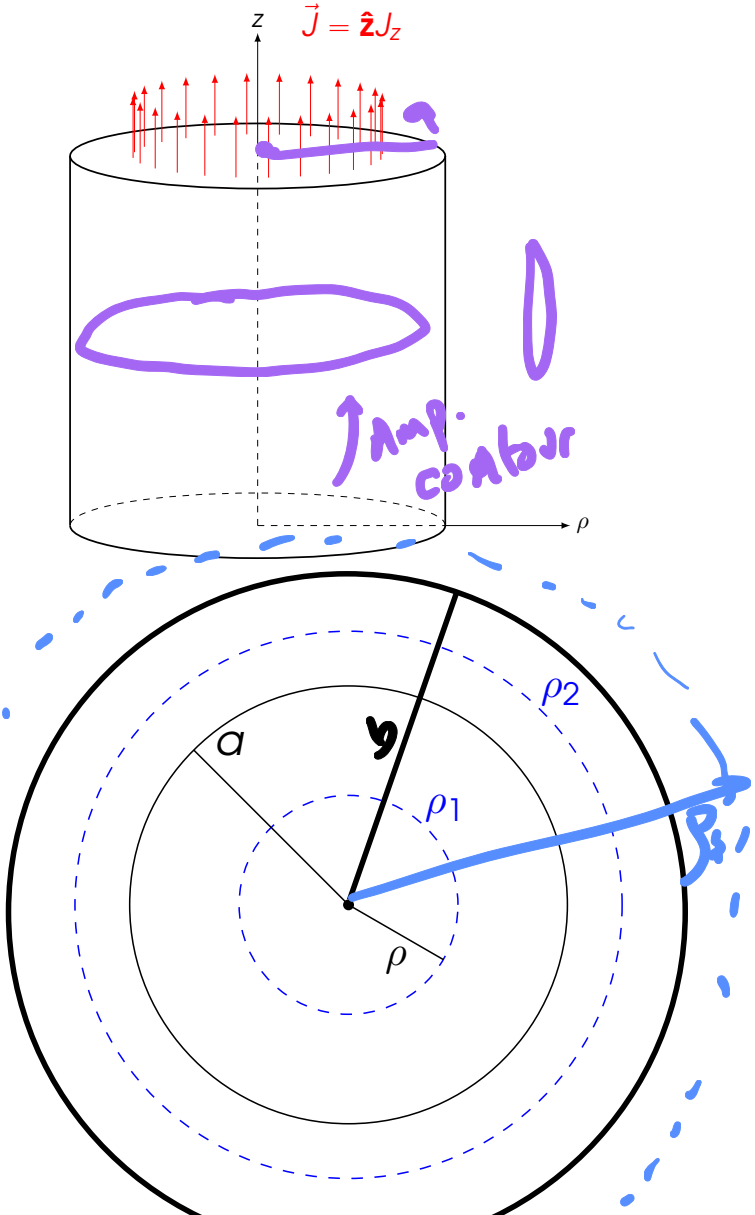


1. Draw a picture, including Ampelian surface
2. $\oint_C \vec{H}\cdot d\vec{\ell} = I_{encl.}$
3. $\vec{H} = \hat{\rho}H_\rho + \hat{\phi}H_\phi + \hat{z}H_z$
4. $d\vec{\ell} = \hat{\phi}\rho\,d\phi = \hat{\phi}b\,d\phi$
5. $\oint \vec{H}\cdot d\vec{\ell} = \int_0^{2\pi} H_\phi b\,d\phi = H_\phi b 2\pi$
6. $I_{encl.} = I$

$H_\phi = \frac{I}{2\pi b}$

$\vec{H} = \hat{\phi}H_\phi = \hat{\phi}\frac{I}{2\pi b}$

AMPERE'S LAW: COAXIAL CABLE



1. Draw a picture, incl. Amp. surface path
2. $\oint_C \vec{H} \cdot d\vec{\ell} = I_{encl}$
3. $\vec{H} = \hat{\rho} H_\rho + \hat{\phi} H_\phi + \hat{z} H_z$
4. $d\vec{\ell} = \hat{\phi}\, \rho\, d\phi$
5. $\oint_C \vec{H} \cdot d\vec{\ell} = \int_0^{2\pi} H_\phi \rho\, d\phi = 2\pi\rho H_\phi$

Ampere's Law: Coaxial Cable

Assuming J_z is uniform

① $I_{encl.} = \oint_S \vec{J} \cdot d\vec{s}$

$= \int_0^{2\pi} \int_0^{\rho'} \hat{z} J_z \cdot \hat{z}\, \rho' d\rho' d\phi$

$= J_z \int_0^{2\pi} d\phi \int_0^{\rho'} \rho' d\rho'$

$= J_z (2\pi) \left(\frac{\rho'^2}{2}\right)$

$\rho' \rightarrow$ "source" current radius

Case 1: $\rho < a$

$\rho_0 =$ ρ'

$H_\phi =$ $\frac{J_z \rho}{2}$

Case 2: $\rho > a$

$\rho_0 =$ a

$H_\phi =$ $\frac{J_z a^2}{2\rho}$

② All Together...

$2\pi\rho H_\phi = J_z \pi \rho_0^2$

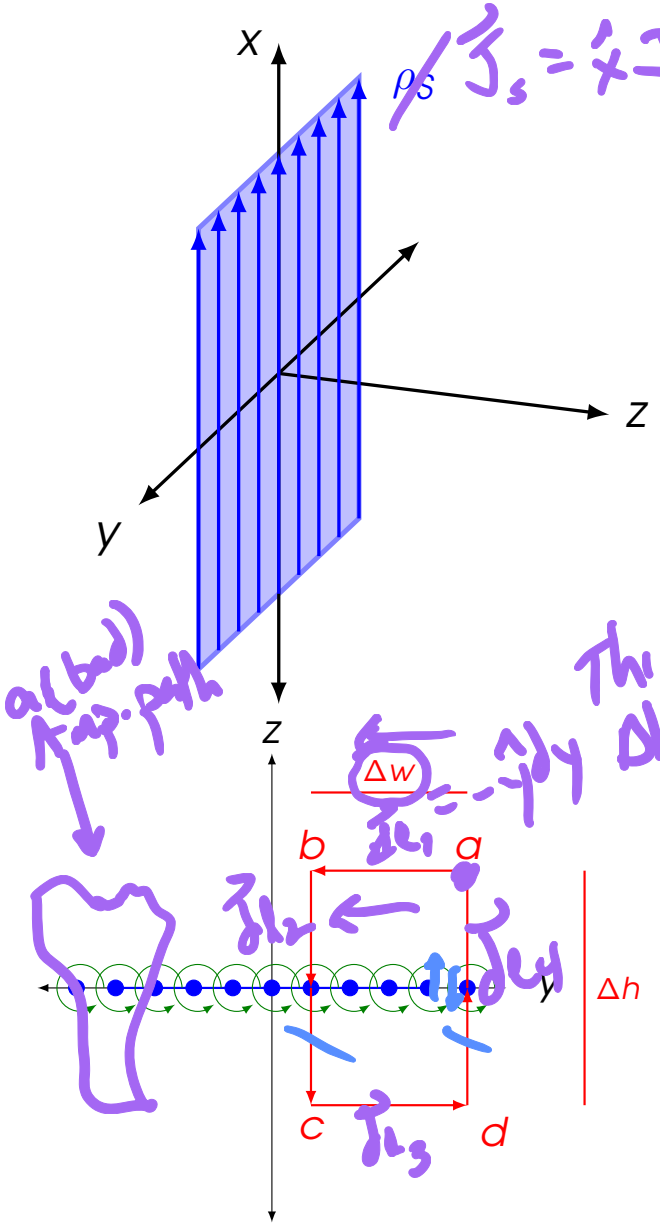
$\vec{H} = \hat{\phi} H_\phi = \hat{\phi} \frac{J_z \rho_0^2}{2\rho}$

Now, plot H_ϕ...



Ampere's Law: Sheet Charge

Current



$\vec{J}_s = \hat{x} J_{sx}$

a(bad) Amp. path

Thin Sheet $\Delta h \to 0$

$\vec{dl}_1 = -\hat{y}\,dy$

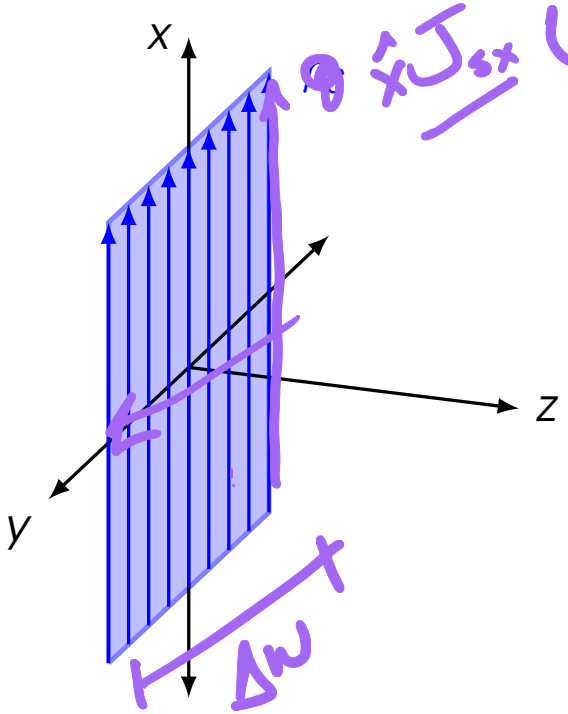
1. Draw a picture, including Amp. path
2. $\oint_C \vec{H}\cdot\vec{dl} = I_{encl.}$
3. $\oint_C \vec{H}\cdot\vec{dl} = \int_a^b \vec{H}\cdot\vec{dl}_1 + \int_b^c \vec{H}\cdot\vec{dl}_2 + \int_c^d \vec{H}\cdot\vec{dl}_3 + \int_d^a \vec{H}\cdot\vec{dl}_4$

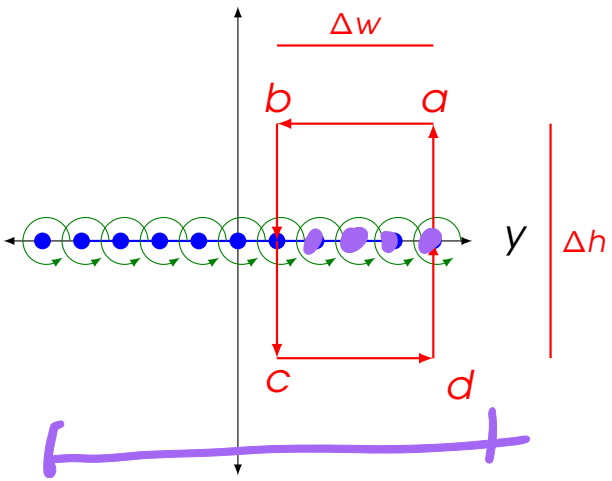
Left side

$= \int_a^b -\hat{y}H_y \cdot (-\hat{y}\,dy) + \int_c^d \hat{y}H_y \cdot \hat{y}\,dy$

$= H_y \Delta w + H_y \Delta w = 2H_y \Delta w$

AMPERE'S LAW: SHEET CHARGE





$\hat{x} J_{sx}$ (A/m)

(6) $I_{encl.} = J_{sx} \Delta w$

Put it all together:

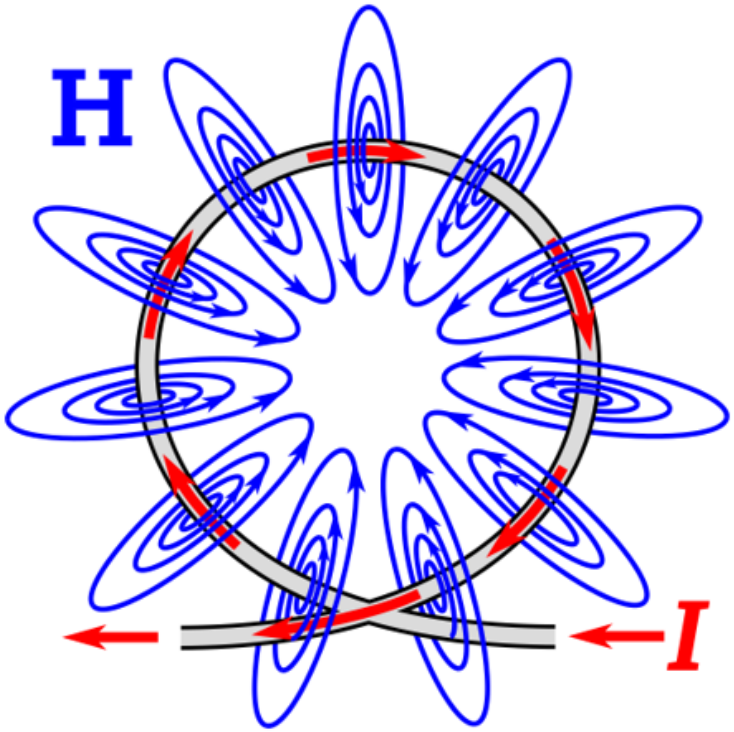
(1) $\oint_C \vec{H} \cdot d\vec{\ell} = I_{encl}$

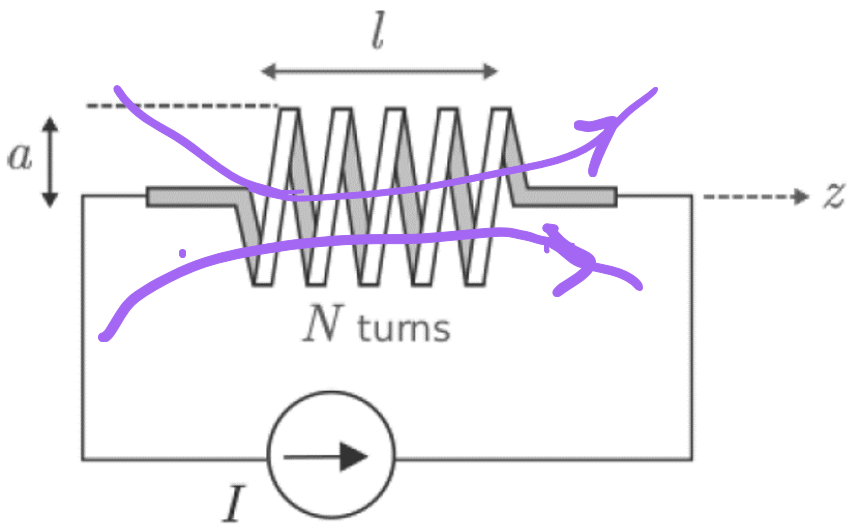
$2 H_y \Delta w = J_{sx} \Delta w$

$$H_y = \frac{J_{sx}}{2}$$

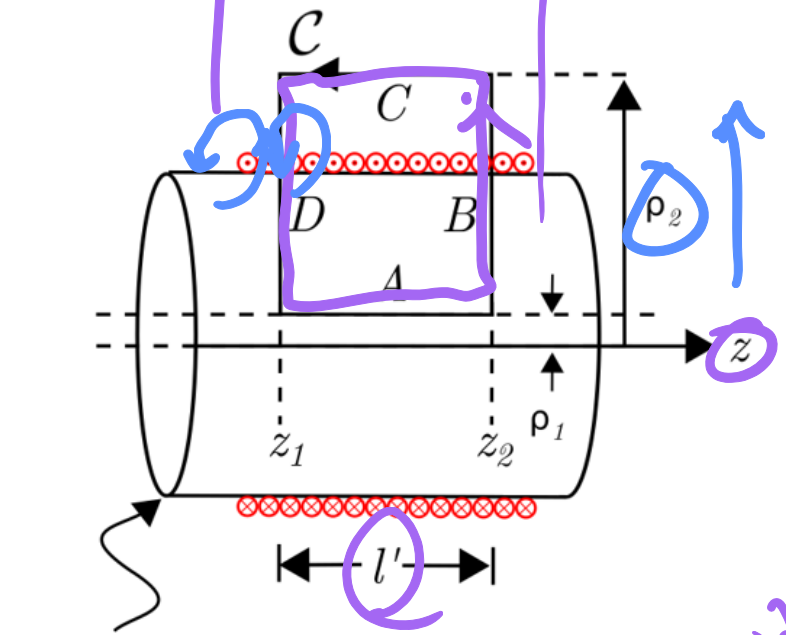
$$\vec{H} = \hat{y} H_y = \hat{y} \frac{J_{sx}}{2}$$

Ampere's Law: Solenoid





Ampere's Law: Solenoid



1. Draw a picture, incl. Amp. path
2. $\oint_C \vec{H}\cdot d\vec{\ell} = I_{encl.}$
3. $\vec{H} = \hat{\rho}H_\rho + \hat{\phi}H_\phi + \hat{z}H_z$
4. $\oint_C \vec{H}\cdot d\vec{\ell} = \int_A \vec{H}\cdot d\vec{\ell}_A + \int_B \vec{H}\cdot d\vec{\ell}_B + \int_C \vec{H}\cdot d\vec{\ell} + \int_D \vec{H}\cdot d\vec{\ell}_D$

$\frac{N}{\ell}$ → winding density

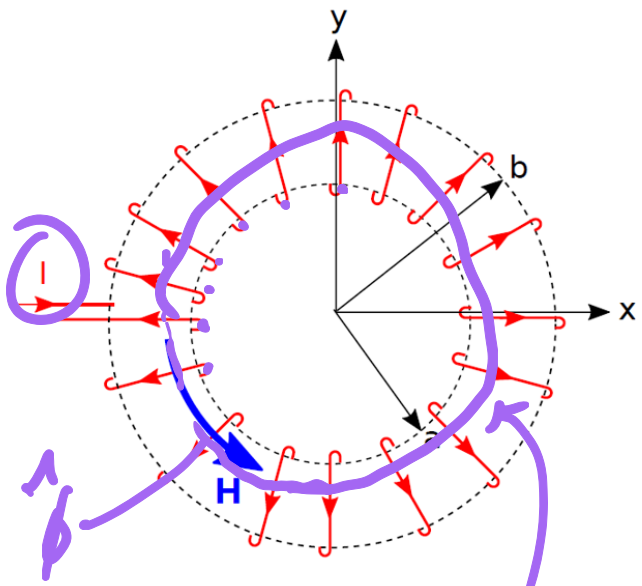
$C \to \infty$ distance

$= \int_A \hat{z}H_z \cdot \hat{z}\,dz = H_z \ell'$

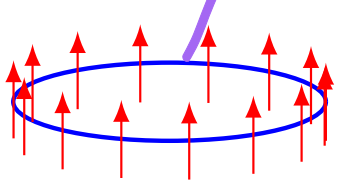
$I_{encl} = \frac{NI}{\ell}\ell'$

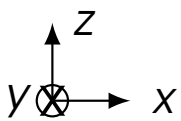
$H_z = \frac{NI}{\ell}$

Ampere's Law: Toroid



Note: the contour of integration is **inside** the core, meaning current through the surface defined by the contour is carried in the windings at the inner edge of the core:





1. Draw a picture, incl. amp. path
2. $\oint_C \vec{H} \cdot d\vec{L} = I_{encl.}$
   $\vec{H} = \hat{\rho} H_\rho + \hat{\phi} H_\phi + \hat{z} H_z$
   $d\vec{L} = \hat{\phi}\, \rho\, d\phi$
3. $\int_0^{2\pi} H_\phi \rho\, d\phi = 2\pi\rho H_\phi$
4. $I_{encl} = IN$
5. $\vec{H} = \hat{\phi} H_\phi = \hat{\phi} \frac{NI}{2\pi\rho}$

Maxwell's Equations: Static Fields

Law	Integral Form	Point Form
Gauss's Law for Electricity	$\oint_S \vec{E}\cdot d\vec{S} = \frac{Q_{\text{encls}}}{\varepsilon}$	$\nabla\cdot\vec{E} = \frac{\rho}{\varepsilon_V}$
Gauss's Law for Magnetism	$\oint_S \vec{B}\cdot d\vec{S} = 0$	$\nabla\cdot\vec{B} = 0$
Faraday's Law	$\oint_\ell \vec{E}\cdot d\vec{\ell} = 0$	$\nabla\times\vec{E} = 0$
Ampere's Law	$\oint_\ell \vec{H}\cdot d\vec{\ell} = I_{\text{encl}}$	$\nabla\times\vec{H} = \vec{J}$